

Edutech Solution – AI & ML Internship

Task 6: Data Cleaning & Preprocessing

Professional GitHub and academic submission report for end-to-end preprocessing pipeline.

Tools Used: Python, Pandas, NumPy, Scikit-learn, Seaborn

Dataset: Telecom Churn / Online Retail

Project Objective

Develop a complete preprocessing system to clean raw datasets for machine learning by handling duplicates, nulls, outliers, encoding, scaling, and transformation.

Step	Description
1	Load and inspect dataset
2	Check duplicates and missing values
3	Apply KNN Imputer
4	Remove outliers using Z-score
5	Encode categorical features
6	Scale numerical features
7	Transform skewed columns
8	Save cleaned dataset

Implementation Code

Import Libraries

```
import pandas as pd
import numpy as np
from sklearn.impute import KNNImputer
from sklearn.preprocessing import StandardScaler, LabelEncoder
from scipy import stats
```

Load Dataset

```
df = pd.read_csv("dataset.csv")
print(df.head())
print(df.info())
```

Check Missing Values

```
print(df.isnull().sum())
print(df.duplicated().sum())
```

KNN Imputation

```
numeric_cols = df.select_dtypes(include=np.number).columns
imputer = KNNImputer(n_neighbors=5)
df[numeric_cols] = imputer.fit_transform(df[numeric_cols])
```

Outlier Removal

```
z_scores = np.abs(stats.zscore(df[numeric_cols]))
df = df[(z_scores < 3).all(axis=1)]
```

Categorical Encoding

```
cat_cols = df.select_dtypes(include='object').columns
encoder = LabelEncoder()
for col in cat_cols:
    df[col] = encoder.fit_transform(df[col].astype(str))
```

Feature Scaling

```
scaler = StandardScaler()
df[numeric_cols] = scaler.fit_transform(df[numeric_cols])
```

Skewness Handling

```
for col in numeric_cols:
    if df[col].skew() > 1:
        df[col] = np.log1p(df[col])
```

Export Dataset

```
df.to_csv("cleaned_dataset.csv", index=False)
```

Interview Questions & Answers

Q: Difference between normalization and standardization?

A: Normalization scales data to a fixed range, while standardization centers around mean 0 with standard deviation 1.

Q: When to use KNN Imputer?

A: When feature relationships matter and contextual imputation is needed.

Q: What is dummy variable trap?

A: Multicollinearity caused by redundant encoded variables.

Q: How do you handle outliers?

A: Using Z-score, IQR, capping, transformation, or removal.

Q: Feature engineering vs preprocessing?

A: Preprocessing cleans data; feature engineering creates useful predictive features.

Final Outcome

A production-ready cleaned dataset and reusable preprocessing pipeline were successfully developed.